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SYSTEMATIC REVIEW

Mental Health in Elite Athletes: A Systematic Review of Suicidal Behaviour as Compared to the General Population

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Abstract

Background and Objective Previous systematic reviews on mental health in athletes have found athletes to be at a potentially increased risk for mental health diagnoses compared to the public. Multiple cross-sectional studies have examined suicide behaviour within different athlete populations, but there is a need for a comprehensive review to synthesize and identify risk factors and epidemiology regarding suicide behaviour in the elite athlete population, especially as it compares to the general population.

Methods A systematic literature search was performed in MEDLINE, EMBASE, Scopus, and Web of Science from 1990 to January 2023. Inclusion criteria included original peer-reviewed research articles examining suicidal ideation, suicide attempt, or suicide completion within elite athlete populations. Exclusion criteria included athletes participating in high-school or Paralympic level sports, studies that did not report results regarding elite athletes and non-athletes separately, and non-peer reviewed work. All studies were screened for inclusion by two independent reviewers. The primary outcome variables extracted from included studies included rates, risk factors, and protective factors for suicide behaviour. The study quality and risk of bias was evaluated for each study using the Joanna-Briggs Institute (JBI) critical appraisal tools.

Results Of the 875 unique studies identified, 22 studies, all of which were cross-sectional in nature, met the inclusion criteria. Seven studies evaluated previous athletes, 13 studies evaluated current athletes, and two studies included a combination of previous and current athletes. Seven studies involved varsity college athletes, nine involved professional athletes of various sports, and six focused on international or Olympic level athletes. The rate of suicidal ideation in professional athletes ranged from 6.9 to 18% across four studies, while the rate in collegiate athletes ranged from 3.7 to 6.5% across three studies. Ten studies compared athletes to the general population, the majority of which found athletes to be at reduced risk of suicidal ideation, suicide attempt, and suicide completion. Only one study found athletes to have increased rates of suicide compared to matched non-athletes. Risk factors for suicide behaviour identified across multiple studies included male sex, non-white race, older age, and depression. Player position, athletic level, sport played, and injuries showed trends of having limited effect on suicide behaviour risk.

Conclusion This review suggests that elite athletes generally demonstrate reduced risk of suicidal ideation, suicide attempt, and suicide completion compared to the general population. Coaches should remain aware of specific factors, such as male sex, non-white race, and higher athletic level, in order to better identify at-risk athletes. Limitations of this review include the heterogeneity in the methodology and athlete populations across the included studies. Therefore, future targeted research is essential to compare suicide behaviour between sports and identify sport-specific suicide risk factors.

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Key Points
Elite athletes demonstrate reduced risk for suicidal behaviour compared to non-athletes
Male sex, non-white race, and higher level of competition may increase risk for suicide in elite athletes
Longitudinal, targeted studies are necessary to better identify specific time periods where elite athletes may be more susceptible to suicidal behaviour

1 Introduction

Suicide is unfortunately a common global phenomenon, resulting in over 800,000 deaths worldwide in 2015, accounting for 1.4% of premature deaths [1, 2]. Suicide currently ranks as the fourth leading cause of death among 15- to 29-year-olds worldwide, which overlaps with the typical age of elite athletes in most sports [2]. Furthermore, suicide appears to be increasing in frequency, as the age-adjusted suicide rate in the United States rose from 10.4 per 100,000 in 2000 to 14.2 per 100,000 in 2018, an increase of 36.7% [3]. As this public health emergency continues to worsen, there has been an emphasis on research to examine the epidemiology of suicide in different populations [1].

Physical exercise has well-documented psychological benefits and is often incorporated in treatment plans for patients with depression, anxiety, or psychotic disorders [4–9]. For this reason, athletes have been perceived as being more immune to the effects of mental health disorders than the general population [10]. However, the current literature on mental illness within athletes is conflicting. A recent review found that participation in sport improves psychological health outcomes [11], yet other reviews have shown that athletes are just as likely or even more likely than the general population to experience mental illness symptoms [12, 13]. A large study across the United States identified that although moderate exercise improves mental health burden, exercising 23 or more times per month or longer than 90 min per session, rates which are often reached by elite athletes, may be associated with worse mental health outcomes [14]. A recent meta-analysis found a prevalence of 34% for symptoms of anxiety/depression in current elite athletes, and 26% in former elite athletes, which is higher than the estimated 20% rate identified within the general public [15–17]. Multiple hypotheses for these trends exist, including stressors unique to high-level sports such as performance expectation, hazing, transitioning from sport, risk-taking tendencies, and sports injuries [18–20]. Athletes tend not to seek support for mental health concerns due to stigma and a perception that seeking help is a sign of weakness, which causes concern for significant manifestations, such as suicide, of mental health issues within this population [21, 22].

Although most research on mental illness in athletes focuses on depression and anxiety symptoms, multiple studies examining suicidal ideation, suicide attempt, or suicide completion within various athlete populations have been performed. The populations in these studies range from college athletes to professional athletes and vary widely in terms of sport, sample size, and method of assessment. Given the heterogeneity in studies and individual outcomes, an overview of these studies and other available evidence about factors surrounding suicide behaviour within athletes is necessary to obtain a more complete understanding of mental health in this population. Therefore, the purpose of this review is to examine risk factors, protective factors, and epidemiology of suicide behaviour within collegiate, professional, or international athletes to help clinicians and coaches better identify high-risk athletes. The research questions this review aims to answer are as follows: How often do elite athletes demonstrate suicidal behaviour and how does that compare to non-athletes? What factors are associated with increased risk for suicidal behaviour within the elite athlete population?

2 Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. This study was registered and is publicly accessible on PROSPERO under the registration number CRD42023395990.

2.1 Literature Search

The literature searches for this project were completed in January 2023. The following databases were searched: Ovid Medline, Ovid Embase, Scopus, and Web of Science. A combination of MeSH (Medical Subject Headings) and keywords was used to complete the search. MeSH terms included: Suicide, Completed; Suicide, Attempted; Suicide; Suicidal Ideation; Athletes and Sports. Keywords included: suicide; suicidal ideation; suicidal thought*; suicidality attempt*; suicide; self-harm; athlete*; athletic* and sports (* indicates truncation of word or phrase). The MeSH terms and counterpart keywords were combined using the Boolean operator 'OR'. The final search strategy is available in Supplementary Table S1 (see the electronic supplementary material). The search was limited to the English language and a publication date range of 1990 to January 2023. These

limitations were made to minimize confounding variables to ensure comparability of results given recent improvements in technology, media access, and social media. The concepts of suicide and athletes were then combined with the Boolean operator of 'AND' resulting in 1271 references. A total of 396 duplicate references were removed, leaving a final total of 875 citations to review to meet inclusion or exclusion criteria.

2.2 Definitions

Suicide behaviour is defined within this paper as any measure of suicidal ideation, suicide attempt, or suicide completion. Elite athlete is defined as any athlete currently competing or having previously competed at the varsity collegiate, professional, national, international, or Olympic levels [23]. Varsity collegiate level athletes includes athletes of all National Collegiate Athletic Association (NCAA) levels, from division I to division III, but does not include athletes competing at the collegiate club level. A non-athlete is defined as anyone that does not fit the definition of 'elite athlete', as defined above.

2.2.1 Study Inclusion

The inclusion criteria were English language, original peer-reviewed research report (any level of evidence), a study population of either active or retired elite athletes, and any form of measurement of suicidal ideation, suicide attempt, or suicide completion within that study population. Exclusion criteria included participants competing at a high-school or non-varsity collegiate level, studies evaluating Paralympic and Special Olympics athletes due to potential confounding variables, studies with a heterogeneous sample of elite athletes and non-athletes without any reporting of those groups separately, and other types of articles such as reviews, case reports, abstracts, and non-peer reviewed work.

The title and abstract of each study were reviewed first for eligibility based on the inclusion and exclusion criteria by two independent reviewers (GS, HS). If the information was unclear based on the title and abstract, the full-text article was examined. Disagreements between reviewers were resolved through a third independent reviewer (VG). Additionally, all included studies were re-screened by the third reviewer (VG).

2.3 Data Extraction and Statistical Analysis

A standardized data extraction template was created evaluating (1) general study information (study title, author, year of publication, and publication journal), (2) subject characteristics (age and sex), (3) sport information (sport played, competition level, and country), (4) presence of a control group, (5) rates of suicide behaviour (suicidal ideation, suicide attempt, suicide completion), (6) risk factors, and (7) protective factors. Data were extracted initially by one author (GS) and re-evaluated for accuracy by a second author (VG). The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

The mean rate for suicidal ideation was calculated by dividing the total number of participants who presented with suicidal ideation by the total number of participants evaluated in those studies. The mean rate for suicide completion was measured in the same manner. All data are available upon reasonable request made to the corresponding author.

2.4 Assessment of Study Quality

The Joanna-Briggs Institute (JBI) critical appraisal tool for cross-sectional studies was used to assess the methodological quality and risk of bias of all included studies. This has been shown to be the preferred tool for cross-sectional studies [24]. Quality was defined as excellent if none of the eight criteria on the JBI checklist were answered 'no', quality was good if only one to two of the JBI checklist criteria were answered 'no', and quality was defined as moderate if three to four of the JBI checklist criteria were answered 'no'.

3 Results

3.1 Study Characteristics

The search strategy identified a total of 1271 studies. A total of 396 duplicate studies were removed, leaving 875 studies to be screened via title and abstract. Title and abstract screening yielded 59 studies, which were reviewed in full text, and 37 of these studies were excluded. Ultimately, 22 studies were included for final analysis [25–46] (Fig. 1).

Seven studies evaluated previous athletes, 13 studies evaluated current athletes, and two studies included a combination of previous and current athletes. Seven studies involved varsity college athletes, nine involved professional athletes of various sports, and six focused on international or Olympic level athletes (Table 1). Thirteen studies were of excellent quality, five were of good quality, and four were of moderate quality.

3.2 Rates

Rates of suicidal ideation, suicide attempt, and suicide completion were measured differently by different studies (Table 1).

Suicidal ideation was measured through self-reported questionnaires asking participants if they had ever considered suicide in seven studies, and rates were found to range from 3.7 to 18.0%. When only including the four studies on

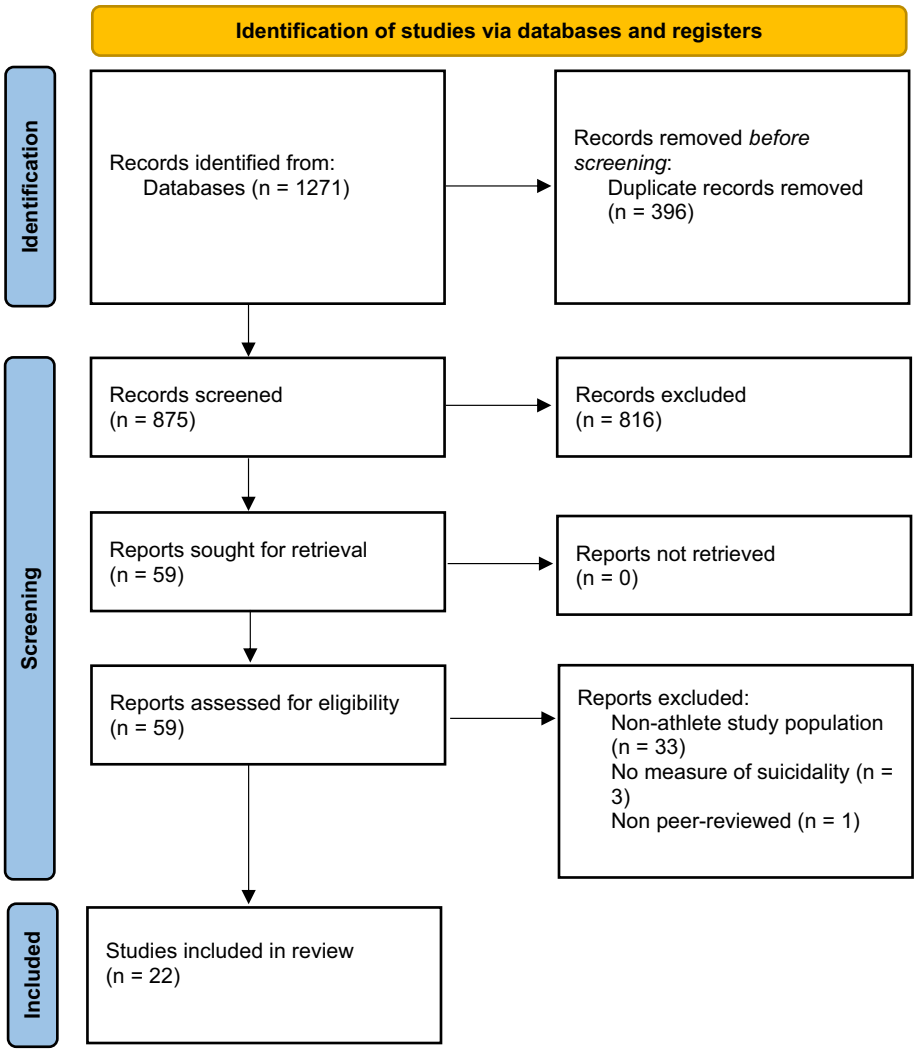

Fig. 1 Study selection flow chart

current professional or international athletes, the rate was found to range from 6.9 to 18% (Table 1) [35, 38, 43, 44]. The average of these studies, accounting for sample sizes, was 14.5%. Among the three studies evaluating collegiate athletes [28, 33, 34], the rate of suicidal ideation was found to range from 3.7 to 6.5%, with an average rate of 4.8%. Only two studies reported rates of suicide attempt, finding rates of 0.76% and 1.3%. Both studies involved varsity NCAA athletes, although the first [34] was a cohort study across two large universities, while the other [28] involved analysis of a national questionnaire administered to college students.

Studies evaluated suicide completion rate by dividing the number of deaths due to suicide by the total number of deaths. Eight studies evaluated suicide completion among a cohort of former or current elite athletes who had died, identifying a range of 0.71–17.0% (Table 1). When only including the six studies on professional or international athletes, the range was 0.7–11.3% and the average was 1.4% [25, 29, 31, 36, 40, 41]. Firearms were found to be the most common method of suicide in studies that reported it. Other, less common, methods included hanging, poisoning, self-laceration, overdose, drowning, intentional motor vehicle accident, and jumping from elevation (Table 1).

3.3 Athlete Versus Non-athlete

Ten studies included a comparison group, comparing athletes to non-athletes (Table 2). Five of these studies evaluated suicidal ideation rates. Three found athletes to be at a significantly reduced risk of suicidal ideation compared to non-athletes [28, 30, 33], while two found no significant difference [32, 34]. Three of these studies reported suicidal ideation rate between groups, all within a collegiate population [28, 33, 34]. In these three studies, the average rate of suicidal ideation among athletes was 4.8%, while the average among non-athletes was 9.1%. One study only reported odds ratios (ORs) for athletes compared to non-athletes for suicidal ideation, finding an OR of 0.574 ($p < 0.01$) (Table 2) [30].

Table 1 Study information and suicidality rates

Study	Sport + competitive level	Country	Sample	Current vs previous athlete	Sex	Suicide ideation rate	Suicide attempt rate	Suicide completion rate	Suicide method	Quality
Abel and Kruger [26]	Professional baseball (MLB)	USA/Canada	Database of 126 suicides within MLB players, managers, umpires, and minor league players	Previous	M	N/A	N/A	N/A	Guns, knives/razors, hanging	Moderate
Alamdarloo et al. [27][a]	Professional athletes across an array of sports	Iran	140 athletes from Shiraz, Iran, engaged in various sports in the country's top leagues	Current	NS	N/A	N/A	N/A	N/A	Good
Anchuri et al. [28]	NCAA (any varsity sport)	USA	20,589 athletes from the NCHA from Fall 2011 to Spring 2015	Current	M + F	6.4%	1.3%	N/A	N/A	Excellent
Duncombe et al. [29]	Olympic (any Olympic sport)	USA	8124 athletes who participated in either summer or winter Olympic Games between 1912 and 2012	Previous	M + F	N/A	N/A	1.3% of all Olympic athlete deaths were due to suicide	Firearms, poisoning	Excellent
Edwards et al. [30][a]	NCAA (any varsity sport)	USA	54,479 athletes from the NCHA from 2011 to 2019	Current	M + F	N/A	N/A	N/A	N/A	Excellent
Gouttebarge et al. [31]	Professional soccer	International	195 deaths with known cause from 2007 to 2013 among active and recently retired professional footballers	Current + previous	M	N/A	N/A	10.3% of deaths were due to suicide	N/A	Moderate
Jovanović et al. [32][a]	International or Olympic (any Olympic sport)	Slovenia	97 athletes from the Olympic Committee of Slovenia database	Current	M + F	Average Personality Assessment Inventory (PAI) score: 2.18	N/A	N/A	N/A	Excellent
Khader et al. [33]	NCAA (any varsity sport)	USA	32,273 from the NCHA from 2011 to 2014	Current	M + F	3.74%	N/A	N/A	N/A	Excellent

Table 1 (continued)

Study	Sport + competitive level	Country	Sample	Current vs previous athlete	Sex	Suicide ideation rate	Suicide attempt rate	Suicide completion rate	Suicide method	Quality
Kokotailo et al. [34]	NCAA (any varsity sport)	USA	217 athletes from two large Midwestern Universities	Current	M + F	5.6%	0.76%	N/A	N/A	Good
Konietzny et al. [35]	International or Olympic (any Olympic sport)	Germany	159 athletes with low back pain recruited through the German Olympic Sports Confederation	Current	M + F	6.9%	N/A	N/A	N/A	Excellent
Lehman et al. [25]	Professional American football (NFL)	USA	3439 athletes with at least 5 playing seasons between 1959 and 1988	Previous	M	N/A	N/A	2.2% of all deaths were due to suicide	N/A	Excellent
Lindqvist et al. [36]	International or Olympic in power sports (wrestling, Olympic lifting, powerlifting, throwing events in track and field)	Sweden	1199 athletes identified as having been ranked 1–10 at least once on official Swedish ranking lists between 1960 and 1979	Previous	M	N/A	N/A	11.6% of all deaths were due to suicide	N/A	Excellent
Maron et al. [37]	NCAA (any varsity sport)	USA	4,052,369 active athletes between 2002 and 2011, deaths identified via the US National Registry of Sudden Death in Athletes	Current	M + F	N/A	N/A	17.0% of all deaths were due to suicide	Guns, hanging, overdose	Excellent
Ojio et al. [38]	Top league rugby	Japan	251 athletes from the Japan Rugby Players' Association	Current	M	7.6%	N/A	N/A	N/A	Excellent
Rao et al. [39]	NCAA (any varsity sport)	USA	3,773,309 individual athlete-participant years between 2003 and 2011 identified through the Student Athlete Ethnicity Report	Current	M + F	N/A	N/A	7.3% of all deaths were due to suicide	Gunshot, hanging, jumping from elevation	Good

Table 1 (continued)

Study	Sport + competitive level	Country	Sample	Current vs previous athlete	Sex	Suicide ideation rate	Suicide attempt rate	Suicide completion rate	Suicide method	Quality
Russell et al. [40]	Professional soccer players (Scottish League)	Scotland	7676 athletes identified through Scottish records	Previous	M	N/A	N/A	1.6% of all deaths were due to suicide	Gunshot, self-laceration, overdose, gassing, drowning, hanging	Excellent
Shah et al. [41]	International cricket players	International	2794 deaths identified between 1877 and 2015	Previous	M	N/A	N/A	0.71% of all deaths were due to suicide	Gunshot, self-laceration, overdose, gassing, drowning, hanging	Moderate
Smith [42][b]	Professional baseball (MLB)	USA/Canada	Database of 126 suicides within MLB players, managers, umpires, and minor league players	Previous	M	N/A	N/A	N/A	N/A	Moderate
Sun et al. [43]	Multiple sports at the top level of competition	China	774 athletes from Chinese national and provincial sports teams	Current	M + F	18%	N/A	N/A	N/A	Excellent
Timpka et al. [44]	International or Olympic track and field athletes	Sweden	192 athletes from the Swedish Athletics Team	Current	M + F	15.6%	N/A	N/A	N/A	Excellent
Tran [45][a]	NCAA (any varsity sport)	USA	39,840 athletes from the NCHA from 2010 to 2015	Current	M + F	N/A	N/A	N/A	N/A	Good
Webner and Iverson [46]	Professional American football (NFL)	USA	26,702 athletes who are currently playing in the NFL or had ever played since 1920	Previous + current	M	N/A	N/A	26 deaths by suicide, no % as a function of total deaths provided	Gunshot, hanging, poison, intentional motor vehicle accident, severed artery	Good

F female, *M* male, *MLB* Major League Baseball, *N/A* not applicable, *NCAA* National Collegiate Athletic Association, *NCHA* National College Health Assessment, *NFL* National Football League, *NS* not specified

[a]Study only assessed risk and protective factors, with no reporting of suicidality rates

[b]Study used the same data as Abel and Kruger [26], using different statistical methods

Table 2 Athlete vs non-athlete comparison studies

Study	Sport + competitive level	Sample size (athletes)	Suicidal ideation (* = $p < 0.05$)	Suicide attempt	Suicide completion
Anchuri et al. [28]	NCAA (any varsity sport)	20,589	Athlete: 6.4% Non-athlete: 11.3%[a]	Athlete: 1.3% Non-athlete: 1.6%[a]	Not assessed
Duncombe et al. [29]	Olympic (any Olympic sport)	8124	Not assessed	Not assessed	Athletes had 0.21 years saved compared to general population[a]
Edwards et al. [30][a]	NCAA (any varsity sport)	54,479	Odds ratio for athletes compared to non-athletes: 0.574[a]	Odds ratio for athletes compared to non-athletes: 0.931[a]	Not assessed
Jovanović et al. [32]	International or Olympic (any Olympic sport)	97	Personality Assessment Inventory (PAI) scores: Athlete: 2.18 Non-athlete: 2.52	Not assessed	Not assessed
Khader et al. [33]	NCAA (any varsity sport)	32,273	Athlete: 3.74% Non-athlete: 5.16%[a]	Not assessed	Not assessed
Kokotailo et al. [34]	NCAA (any varsity sport)	217	Athlete: 5.6% Non-athlete: 6.9%	Athlete: 0.76% Non-athlete: 0.89%	Not assessed
Lehman et al. [25]	Professional American football (NFL)	3439	Not assessed	Not assessed	Athletes: 12 suicide deaths Non-athlete matched general population: 25.6 deaths expected[a]
Lindqvist et al. [36]	International or Olympic in power sports (wrestling, Olympic lifting, powerlifting, throwing events in track and field)	1199	Not assessed	Not assessed	Athletes: 21 suicide deaths Non-athlete matched general population: 12.1 deaths expected[b]
Maron et al. [37]	NCAA (any varsity sport)	4,052,369	Not assessed	Not assessed	Athletes: 0.8 deaths/100,000 athlete participation years Non-athlete: 9.8 deaths/100,000 person years[a]
Russell et al. [40]	Professional soccer players (Scottish League)	7676	Not assessed	Not assessed	Athlete: 0.25% Non-athlete: 0.40%

NCAA National Collegiate Athletic Association, *NFL* National Football League

[a]Significant difference favouring athletes

[b]Significant difference favouring non-athletes

Three studies compared suicide attempt rates (Table 2) [28, 30, 34]. Two studies found athletes to be at a significantly reduced risk of suicide attempt compared to non-athletes [28, 30], while one found no significant difference [34]. Two studies reported rates of suicide attempt, identifying rates of 1.3% and 0.8% in athletes, and rates of 1.6% and 0.9% in non-athletes, respectively [28, 34]. Five studies compared suicide completion rates in athletes to non-athletes (Table 2) [25, 29, 36, 37, 40]. Three studies found athletes to have reduced rates of death by suicide compared to non-athletes [25, 29, 37], one found no significant difference [40], while one found an increased rate of death by suicide [36]. The latter study found mortality from suicide to be increased 2–4 times among former elite Swedish athletes compared to non-athletes.

3.4 Risk and Protective Factors

An array of risk factors for suicide within athlete populations were identified (Table 3). Interestingly, three studies found male athletes to be at increased risk of suicide completion compared to females [28, 37, 39], while one study

Table 3 Risk factors for suicidality

Study			Risk factors								
Study	Sport + competitive level	Sample size (athletes)	Sex	Race	Age	Sport	Athletic level	Position	Depression	Social dysfunction	Miscellaneous/other
Jovanović et al. [32]	International or Olympic (any Olympic sport)	97	NSD on SI			NSD on SI					
Konietzny et al. [35]	International or Olympic (any Olympic sport)	159	NSD on SI		NSD on SI						NSD on SI with level of pain-related thought suppression
Maron et al. [37]	NCAA (any varsity sport)	4,052,369	Male SC risk greater than female	NSD on SC							
Rao et al. [39]	NCAA (any varsity sport)	3,773,309 athlete participant seasons	Male SC risk greater than female	NSD on SC		Football increased rate for SC compared to other sports	NSD on SC when comparing D1 to D2 to D3 athletes				
Sun et al. [43]	Multiple sports at the top level of competition	774	Female SC risk greater than male		Younger athletes have greater risk for SI		Elite athletes (ranked within the top 8) had increased risk of SI compared to athletes ranked outside the top 8		Depression predicts SI risk		

Table 3 (continued)

Study			Risk factors								
Study	Sport + competitive level	Sample size (athletes)	Sex	Race	Age	Sport	Athletic level	Position	Depression	Social dysfunction	Miscellaneous/other
Anchuri et al. [28]	NCAA (any varsity sport)	20,589	Male SI and SA risk greater than female	White athletes have reduced SI and SA risk compared to non-white	Older athletes have decreased risk for SI or SA					Increases risk for SI and SA	Illicit substance use, physical assault victim, and sexual assault victim were associated with increased risk of SI and SA Being heterosexual was associated with decreased SI and SA risk
Lehman et al. [25]	Professional American football (NFL)	3439		NSD on SC				NSD on SC between speed and non-speed (defensive + offensive line) positions			
Timpka et al. [44]	International or Olympic track and field athletes	192				NSD on SI between endurance and non-endurance athletes					Retirement, sports injuries, training load, financially disadvantaged childhood has NSD on SI Sexual abuse and lower sense of coherence increases SI risk in women Behavioural disengagement for coping increases SI risk in men

Table 3 (continued)

Study			Risk factors								
Study	Sport + competitive level	Sample size (athletes)	Sex	Race	Age	Sport	Athletic level	Position	Depression	Social dysfunction	Miscellaneous/other
Tran [45]	NCAA (any varsity sport)	39,840		Multiracial, Hispanic, Asian/Pacific Islander, Native American, or black race had greater risk for SI or SA compared to white athletes	Older athletes have greater risk for SI or SA						Higher GPA, part-time enrolment, and attending a 2-year institution all associated with increased risk for SI or SA
Lindqvist et al. [36]	International or Olympic in power sports (wrestling, Olympic lifting, powerlifting, throwing events in track and field)	1199			Increased risk for SC at age 30–50						
Duncombe et al. [29]	Olympic (any Olympic sport)	8124				Rowing, swimming, and athletics have decreased SC risk compared to fencing and shooting	NSD on SC when comparing medallists to non-medallists				
Russell et al. [40]	Professional soccer players (Scottish League)	7676						NSD on SC between different positions on soccer team			

Table 3 (continued)

Study			Risk factors									
Study	Sport + competitive level	Sample size (athletes)	Sex	Race	Age	Sport	Athletic level	Position	Depression	Social dysfunction	Miscellaneous/other
Alamdarloo et al. [27]	Professional athletes across an array of sports	140							Severe depression predicts SI risk	NSD on SI	Anxiety and somatic symptoms have NSD on SI

D Division, *GPA* Grade Point Average, *NCAA* National Collegiate Athletic Association, *NFL* National Football League, *NSD* no significant difference or effect, *SA* suicide attempt, *SC* suicide completion, *SI* suicidal ideation

found females to be at greater risk [43]. Two other studies found no correlation with sex and suicidal ideation [32, 35] (Fig. 2). Similarly, the risk factor of age varied between studies. One study identified athletes aged 30–50, which the study identified as the late-stages of one's career or early retirement, to be the age with the greatest risk for suicide completion [36]. Another study identified younger athletes to be at greatest risk for suicidal ideation[43], while two other studies found older age to increase risk of both suicidal ideation and suicide attempt [28, 45] (Fig. 2). One study of collegiate athletes reported that identifying as multiracial, Hispanic, Asian/Pacific Islander, Native American, or black was associated with an increased risk for suicidal ideation and suicide attempt in college athletes, while identifying as white was associated with decreased risk [45]. Similarly, a study of collegiate athletes found white athletes had a reduced risk of suicidal ideation and suicide attempt compared to non-white athletes [28]. However, two separate studies on college athletes found no significant difference in suicide completion based on race [37, 39]. Similarly, a study of National Football League (NFL) athletes found no significant difference in suicide completion between black and white athletes (Table 3) [25].

A study of Chinese athletes found that higher skill-level athletes, as defined by athletes ranked nationally within the top 8 for their specific sport, had an increased risk for suicidal ideation [43]. Alternatively, a study of NCAA athletes found no significant difference in suicide completion based on NCAA division (division I vs division II vs division III) [39]. Similarly, a study of Olympic athletes found no significant difference in suicide completion between medallists and non-medallists (Table 3) [29].

Differences were found between sport and position as well. Two studies identified no difference in suicidal ideation between athletes of different sports [32, 44]. Alternatively, one study found American football athletes to be at an increased rate for suicide completion compared to other sports [39]. Similarly, a study of Olympic athletes found rowing, swimming, and athletics participants to have decreased suicide completion risk compared to fencing and shooting athletes [29]. Two studies examined player position within professional soccer and NFL athletes, finding no significant difference for suicide completion based on position [25, 40].

Other risk factors for suicide behaviour found by various studies include illicit substance use [28], history of experiencing sexual assault or physical assault [28], higher grade point average (GPA) [45], part-time University enrolment [45], and behavioural disengagement for coping [44]. Notably, retirement, sports injuries, training load, and pain were not associated with any increased risk of suicide behaviour. One study found self-reported difficulty with social relationships to have a significant correlation

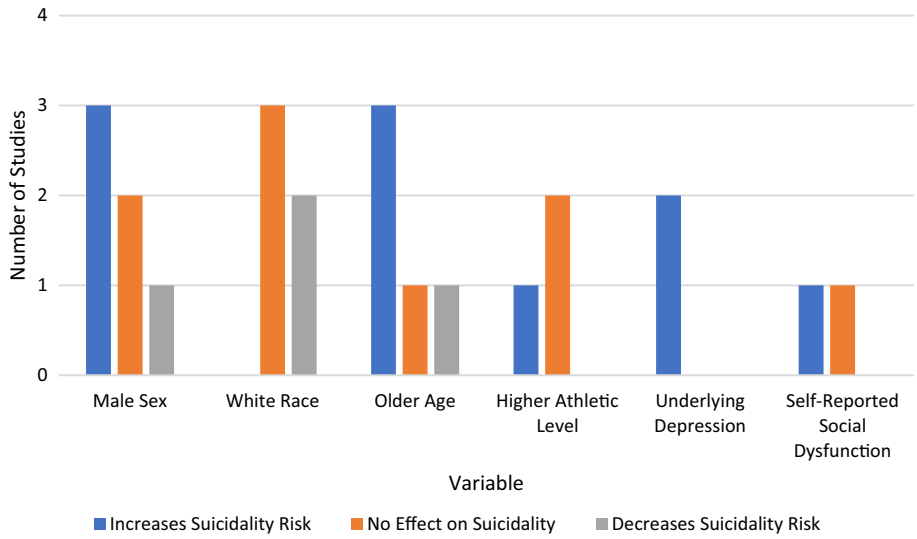

Fig. 2 Risk factors evaluated in more than one study

with suicide attempt in athletes [28], while another study found no significant correlation between social dysfunction and suicide risk [27]. Two studies evaluated underlying depression as a risk factor for suicidal ideation, both finding a significant correlation (Fig. 2) [27, 43].

One study compared athletes with suicidal ideation to athletes without suicidal ideation, finding that players with suicidal ideation reported greater changes in appetite, changes in weight, sleep problems, reduced competitiveness, and thinking about a career after retirement compared to those without suicidal ideation [38].

4 Discussion

This study systematically analysed and synthesized information from studies evaluating suicide behaviour within elite athlete populations. The primary aim of this review was to describe the published evidence on the epidemiology, risk factors, and protective factors for suicide behaviour within elite athlete populations. The major findings of this review, which included 22 studies, suggest that there is heterogeneity in terms of universal risk for suicide behaviour and individual risk factors associated with suicide behaviour within elite athlete populations. However, a notable trend was that athletes are at reduced risk for suicide behaviour compared to matched non-athlete populations.

Previous literature examining mental health symptoms and disorders in elite athletes has demonstrated conflicting findings. A recent meta-analysis identified a prevalence of 34% for anxiety/depression in current elite athletes and 26% in former elite athletes [15]. These rates are slightly higher than general population studies, where the prevalence of anxiety/depression tends to be under 25% [16, 17, 47]. A different systematic review found major depressive disorder to be no more likely in athletes than non-athletes [22], while another systematic review found comparable levels of anxiety between elite athletes and the general population [48]. The current study also identified broad and conflicting results in terms of suicide behaviour within athletes. When considering the ten studies that included a control group of non-athletes or the matched general public for any measure of suicide behaviour, six studies found that athletes have significantly less risk than non-athletes, three found no significant difference between athletes and non-athletes, while one found athletes to be at greater risk than non-athletes. Therefore, this review identifies a trend that suicide behaviour, as measured by suicidal ideation, suicide attempt, or suicide completion, is lower in elite athletes compared to non-athletes. This aligns with the findings of a recent narrative review by Pichler et al. [20]. It is important to note that the one study that found an increased risk for suicide completion examined a cohort of athletes who participated in competition prior to the banning of anabolic androgenic steroids (AAS) [36]. The authors did not formally assess AAS use in this cohort but hypothesized that about one third of the athletes may have used AAS. Previous studies have found AAS users to have an increased risk for major mood disorders and dependence, which can drive suicidal behaviour, and could potentially explain the increased rate found in this study [49–51].

Most of the included studies in this systematic review evaluated lifetime prevalence of suicide behaviour. Meta-analyses of lifetime suicidal ideation rates in the general public have found rates of around 9–10% [52, 53], although this can vary significantly by country. In the current study, a pooled suicidal ideation rate of 14.5% was found in studies evaluating professional or international athletes, while a rate of 4.8% was found across three studies evaluating collegiate athletes. One study evaluating international

athletes identified a suicidal ideation rate of 18%, and this study had a sample size more than three times greater than the next largest study of professional athletes, thus shifting the pooled rate vastly [43]. Therefore, although the average suicidal ideation rate among professional/international athletes was found to be larger than the general public rate of 9–10%, it is important to consider how the majority of studies individually identified rates less than the general public rate. Meta-analyses of lifetime suicide attempt rates in the general public have identified a rate of around 2.7% [52]. Only two studies in this review evaluated suicide attempt rate, both of which evaluated collegiate populations and included a non-athlete comparison group. Those studies identified rates of 0.76% and 1.3%, both of which were less than the non-athlete comparison group and the general public rate of 2.7% [28, 34]. This further demonstrates the generalized trend that suicide behaviour is decreased within elite athletes compared to non-athletes, although further research, especially in the realm of suicide ideation, is warranted.

Suicide behaviour rates differed vastly between professional/international athlete and collegiate athlete populations. Interestingly, all three studies evaluating suicidal ideation rates in collegiate athletes found a rate (3.7%, 5.6%, 6.4%) smaller than the lowest rate (6.9%) found within professional/international athletes. The pooled average suicidal ideation rate among collegiate athletes (4.8%) was substantially lower than the average among professional/international athletes (14.5%). No study measured suicide attempts in professional/international athletes, so no comparisons could be made. Further, all the studies that measured suicide completion in professional/international athletes included previous athletes, while the studies on collegiate athletes only included current athletes, so comparisons could not reliably be made. However, one study of professional soccer players did examine strictly current and recently retired players, finding a rate of 10.3% [31], which was between the rates found in the two studies on current college athletes (7.3%, 17.0%). These results are surprising given college athletes are typically associated with having a greater number of risk factors associated with suicide risk, such as financial hardships, increased stress due to juggling sports and academics, and concurrent psychiatric disorders [54–56]. Bruner et al. examined the transition from amateur to elite sport in rookie ice hockey players, finding that these rookie athletes struggled with self-confidence due to the higher level of play and had decreased social support secondary to moving to a new city with little control [57]. Another explanation could be that professional athletes receive greater social scrutiny, due to increased viewership and media attention, which can cause undue stresss [58, 59]. Further research comparing professional athletes to collegiate athletes is necessary to better understand these differences and to implement effective interventions.

This review identified trends that non-white race and male sex may be associated with increased rates of suicide behaviour within elite athletes. Research on the general public also finds males to have higher rates of suicide compared to females, but finds white and Native American individuals to have higher rates of suicide behaviour compared to other races [60–62]. Of the five studies that evaluated race among athletes, three found no significant difference between races and two found white athletes to have reduced suicidal ideation and suicide attempt rates compared to non-white athletes. Notably, both of these studies were performed on collegiate populations. Limited research has been performed on the general college population, but one study did identify white and racial-ethnic minority college students to as exhibiting similar rates of suicidal ideation [63]. The decreased suicide behaviour risk in white athletes, in contradiction to the increased or unchanged risk generally seen in white non-athletes, could potentially be attributed to increased social support and resources available to white athletes [64, 65]. Similarly, a study by Melendez found that black American football athletes at a predominantly white university felt isolated, rejected, and mistrustful of not only their white teammates but also of their non-athlete black classmates [66]. Future research should also examine the potential role of stereotype threat and othering in sports leading to this finding [67].

Only one study within this review compared American football athletes with athletes of other sports, finding football athletes to be at greater risk of suicide behaviour. One hypothesis for this finding is that American football athletes tend to suffer from more concussions than non-football athletes. A recent meta-analysis supports this hypothesis, having found a twofold higher risk of subsequent suicide among patients diagnosed with a concussion compared to individuals without a concussion [68]. Similarly, there is a multitude of literature highlighting a potential association between chronic traumatic encephalopathy (CTE), which American football players are at increased risk for, and suicide behaviour [69–71].

This review also identifies trends that sport position and athletic level may have an effect on suicide behaviour. One study in this review found higher ranked athletes to have increased risk of suicidal ideation compared to lower ranked athletes [43], which contradicts a previous study that found second league female soccer players to have significantly more depression symptoms compared to first league players [72]. Different levels of competition are associated with different stressors, such as financial hardship at lower levels and public scrutiny at higher levels, so correlations could be expected. Many risk and protective factors for suicide behaviour within elite athletes were identified in the

studies included in the review, but no unanimous findings were found across the studies. This aligns with a previous statistical study that found no variable fully mediated the relationship between sport participation and suicidal ideation [73]. Additionally, there was limited examination of variables typically considered unique to athletes that have previously been associated with mental health symptoms, such as sports injury, retirement, illicit substance use, risk taking tendencies, and poor sports performance [15, 74–77]. The two studies that examined sports injuries or sports-related pain found no correlation with suicidal ideation [35, 44]. One study evaluated retirement, also finding no correlation [44]. Retirement from elite sport has previously been linked to increased rates of depression, shifts in identity, and a loss of social support [78, 79]. Given these are all risk factors for suicide, it is crucial for future studies to also comprehensively examine changes in suicidal behaviour during the retirement period [54, 80].

Despite most included studies being of excellent or good quality, all included studies were cross-sectional in nature. This emphasizes the need for longitudinal research examining athletes stratified by sport and competitive level to granularly discern differences. Longitudinal studies have previously been utilized to examine changes in depressive symptoms, anxiety, sleep quality, and motivation over the course of athletic seasons [81, 82]. Therefore, longitudinal studies may better identify specific times in an elite athlete's life, such as during injury or following retirement, where they may be at increased risk for suicidal behaviour. Future longitudinal studies should aim to assess a cohort of elite athletes over the course of at least two seasons, include a matched cohort of non-athletes, and include regular assessments of suicidal behaviour during that course.

This review is not without limitations. A major limitation of this study is the heterogeneity in methodology of each of the studies. Different studies used different questionnaires, definitions, and statistical methods, making analysis and comparison difficult. This led to a wide variety in the patient populations of the included studies but was necessary to achieve an adequate number of studies for review given that no more than three studies evaluated the same sport and competition level. This may create an unintentional bias, as athletes from different sports may experience sport-specific stressors that this review could not fully account for. This study only included English studies and studies published after 1990, which may have led to the exclusion of applicable studies. This may create an unintentional bias by excluding important and relevant information. Further, the definition of 'elite athlete' is subjective. This study defined elite athletes as those competing at a professional, international, or varsity collegiate level, but even within those categories there remains a breadth of various competition levels. Further, this led to the exclusion of studies evaluating high school and club collegiate athletes, potentially resulting in unidentified correlations and rates. Additionally, the inclusion of studies from many different countries resulted in a variable of cultural and societal differences, which could not be accounted for in comparisons. Most included studies examined risk factors associated with suicide, but there was little evaluation of protective factors for elite athletes. As such, it is crucial for future research to examine factors protective against suicide within the elite athlete population so that sports organisations and bodies may generate evidence-based structures to support athletes.

5 Conclusion

This review suggests that elite athletes generally demonstrate reduced risk of suicidal ideation, suicide attempt, and suicide completion compared to non-athletes and the general public. Relevant risk factors for suicidality that should be considered within elite athletes include male sex and non-white race. Age, sport, and athletic level may also play a role in evaluating an athlete's risk profile for suicide. Further high-quality research is needed to better understand how race, athletic level, sports-related injuries, retirement, and substance use may affect suicide behaviour risk within this population. This information may ultimately help identify at-risk athletes, and allow for the allocation of proper resources and support towards mental health and suicide prevention among athlete populations who are most at risk.

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Declarations

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Ethics approval Given the systematic review nature of this article, institutional review board approval was not necessary to obtain.

Informed consent Not applicable given the systematic review nature of this article.

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